

Symbolic Regression for Precision LHC Physics

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Introduction

- In high energy physics, fundamental interactions can be described by the *Standard Model* (SM) of particle physics to a good extent.
- The SM is tested at the world's most powerful microscope: the *Large Hadron Collider* (LHC). Theoretical predictions and observables are compared.



- Formulas are the actual language of theoretical frameworks [1].
- The question:** Can simple, accurate formulas be derived from noisy, complex, collider data?

Methodology

- We use *symbolic regression* (SR) with the PySR [2] library.
- Formulas are represented as expression trees, as in Fig. 1, with an evolutionary algorithm using crossover and mutations, as in Fig. 2, to generate new trees.

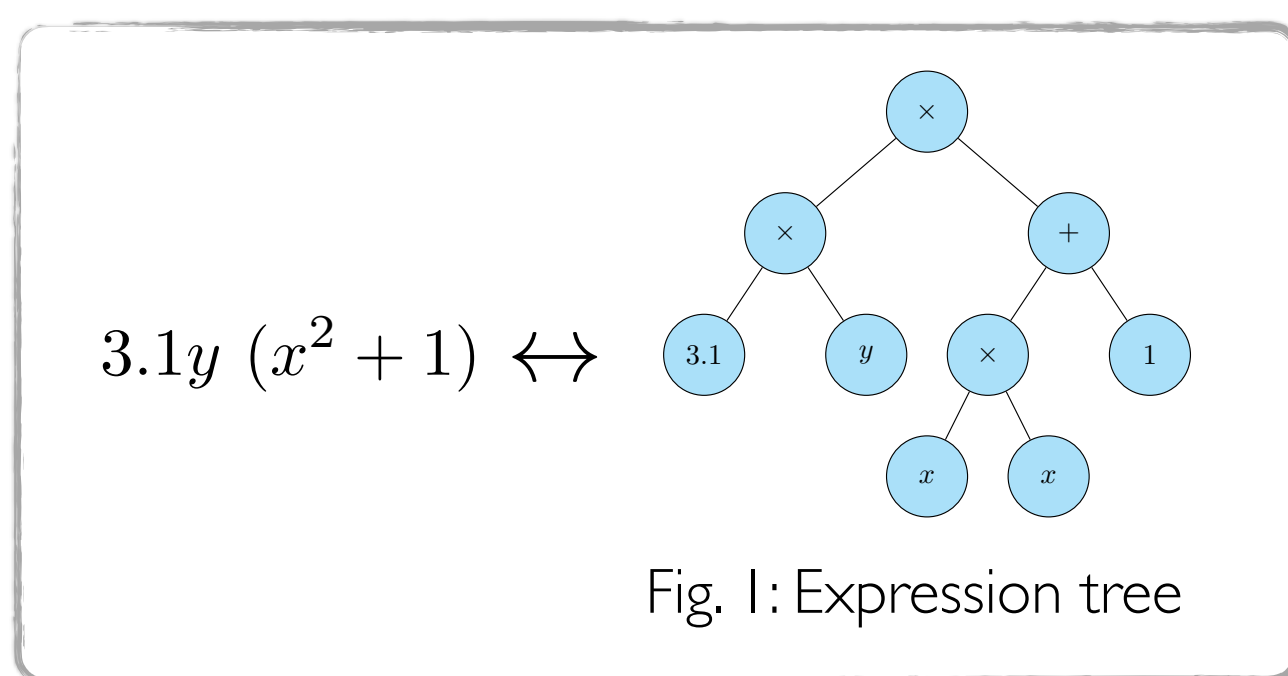


Fig. 1: Expression tree

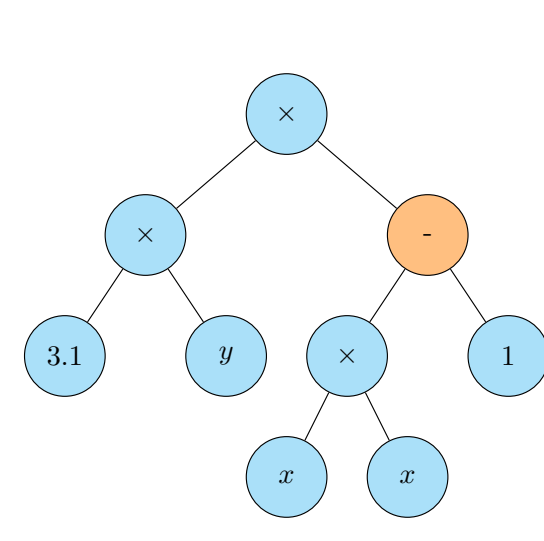


Fig. 2: Mutation

- Each tree has a complexity c , related to its number of nodes.
- Trees can be selected to optimise one of three metrics:

$$L = \frac{1}{n} \sum_{i=1}^n (Y_i - \hat{Y}_i)^2 \quad \text{Minimise} \quad \text{Accuracy:}$$

$$-\frac{\partial \log(L)}{\partial c} \quad \text{Maximise} \quad \text{Score:}$$

$$L \leq 1.5 \times L_{\min} \quad \text{Highest score with} \quad \text{Best:}$$

- We benchmark **equation recovery** from known laws and explore **equation discovery** to describe physical quantities simply and accurately, where no formulas are known.

Results

- Equation recovery** in quantum electrodynamics:

$$e^- e^+ \rightarrow \gamma^* \rightarrow \mu^- \mu^+ \quad \longrightarrow \quad \frac{d\sigma}{d\cos\theta} = \frac{\pi\alpha^2}{2s} (1 + \cos^2\theta)$$

- Under different resolutions we obtain Tab. I. and Fig. 3.

Bins	Accuracy	Best	Score
10	$c_\theta(c_\theta + 0.00798)(0.00111 \cdot c_\theta^2 + 0.03459) + 0.03503$	$c_\theta^2(0.00111 \cdot c_\theta + 0.03459) + 0.03503$	$0.03459 \cdot c_\theta^2 + 0.03503$
20	$c_\theta(c_\theta + 0.01825)(-0.00155 \cdot c_\theta(c_\theta - 0.05138) + 0.03579) + 0.03485$	$c_\theta(0.03447 \cdot c_\theta + 0.00064) + 0.03498$	$0.03447 \cdot c_\theta^2 + 0.03498$
200	$c_\theta^2(-0.64647 \cdot c_\theta(0.00119 \cdot c_\theta - 0.00151) + 0.03495) + 0.03495$	$0.03447 \cdot c_\theta^2 + 0.03495$	$0.03447 \cdot c_\theta^2 + 0.03495$

Tab. I: SR formulas for QED, $c_\theta \equiv \cos\theta$.

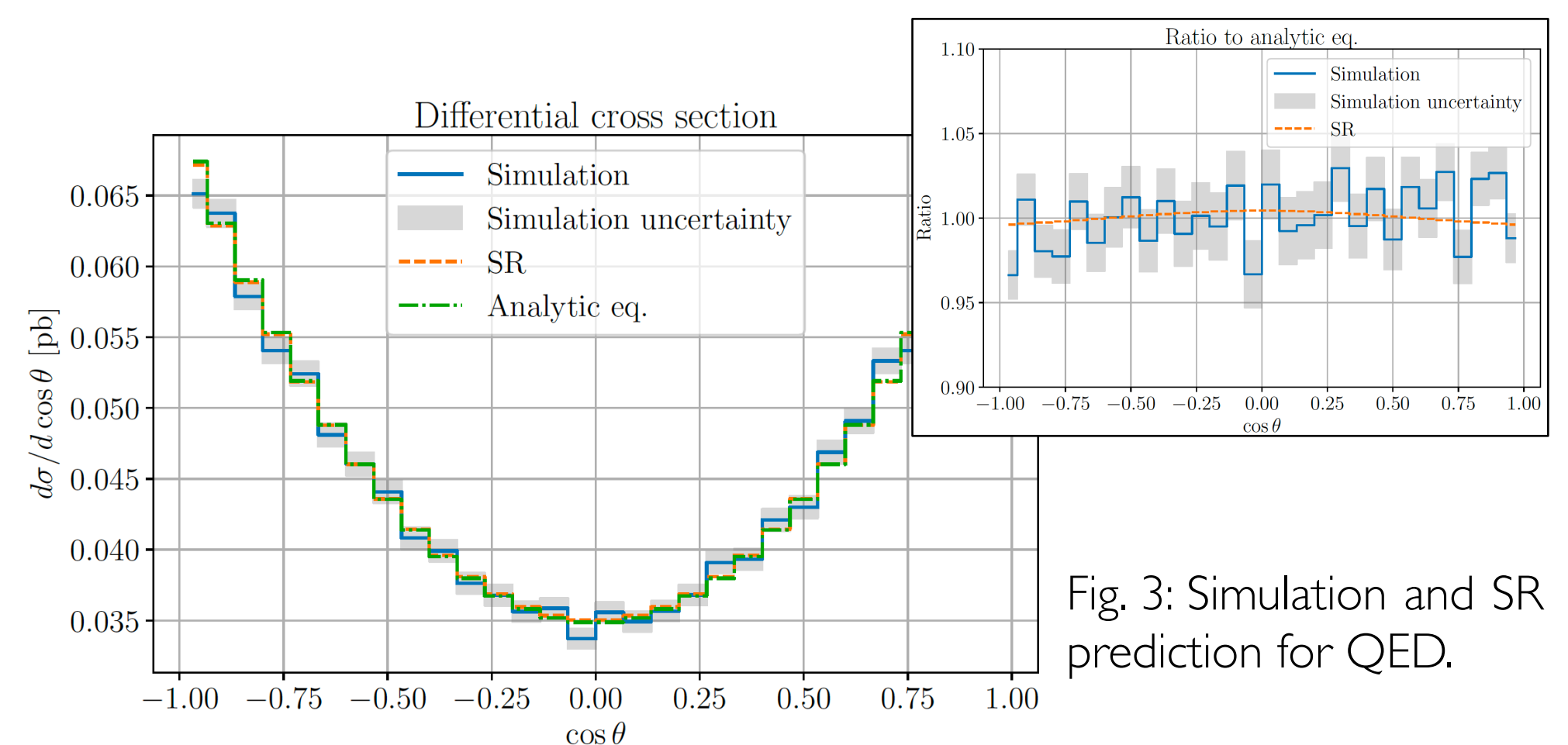


Fig. 3: Simulation and SR prediction for QED.

- Equation discovery** in structure functions (SFs) [3]:

$$p p \rightarrow \gamma^* \rightarrow \mu^- \mu^+ \quad \longrightarrow \quad \frac{d^2\sigma}{dM dy} = \frac{8\pi\alpha^2}{9sM} F(M, y) \quad ?$$

- We obtain smooth and simple formulas for the SFs, as shown in Fig. 4 and Table 2.

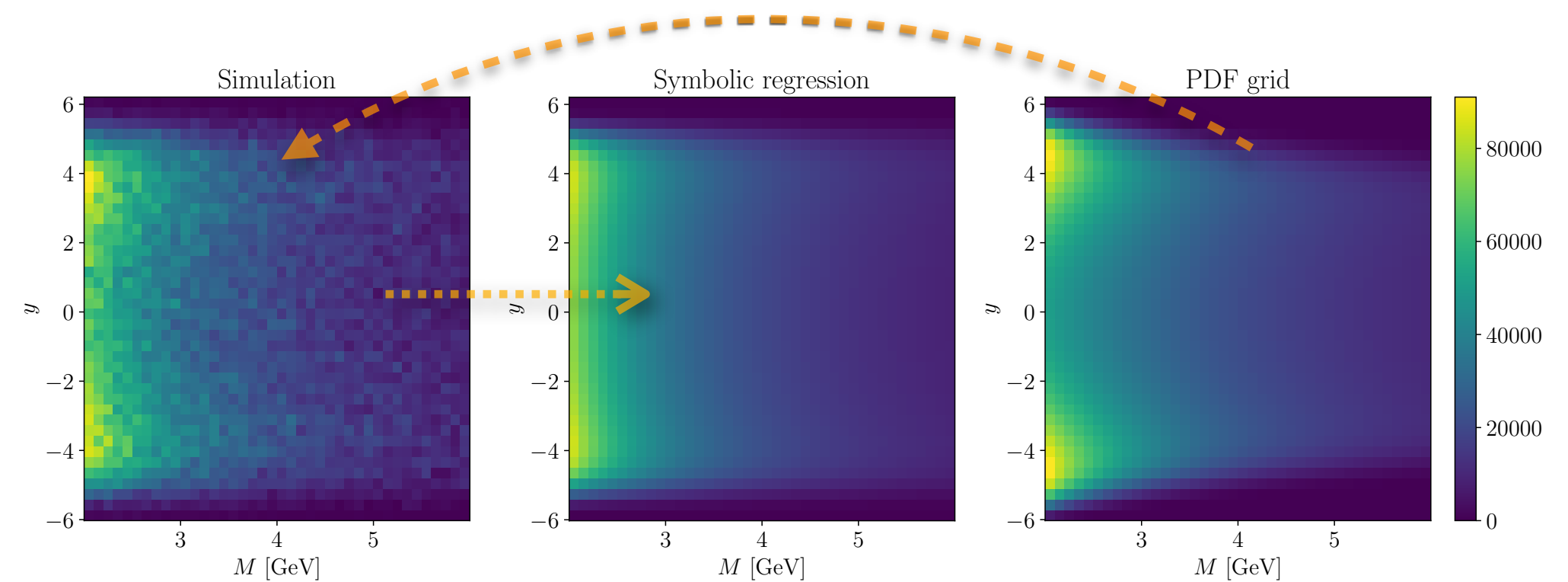


Fig. 4: Simulation, SR fit, and underlying SF (PDF grid).

Complexity	Equation	Score
3	$\frac{9.16 \cdot 10^4}{M}$	0.359
33	$\frac{2.86 \cdot 10^5 \left((0.0461 \cdot 1.15y^2)^{-0.0250-1.15y^2} \right)}{M^2}$	0.387
35	$\frac{2.86 \cdot 10^5 \left((0.0461 \cdot 1.15y^2)^{-0.0250-1.15y^2} \right)}{M^2 + 0.117}$	0.00967

Tab. 2: Selection of SR equations that describe the SF.

- SR expressions improve the description of the data, as seen in Tab. 3.

Metric	Reweighted simulation	Symbolic regression
Root mean square error (RMSE)	6.09×10^3	5.72×10^3
Mean absolute error (MAE)	4.26×10^3	3.74×10^3
Coef. of determination (R^2)	0.8898	0.9030

Tab. 3: Metrics of the simulation and SR compared to PDF baseline.

The answer:

SR can combine the power of machine learning with the intuition of analytical expressions to describe collider data.